

MONASH University

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HONOURS THESIS

Slowly Decaying Correlation in a Simple Metastable Quanutm System

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Abstract

Metastability in open quantum systems is the occurrence of a very slow decaying or stationary state which is not the final stationary state of the system. In this thesis, we present a simple three-body collision model called the partial swap model and investigate the metastable characteristics of this system. Initially, metastability was only observed in the generator spectrum of modified partial swap models with interactions between their environments. However, when perturbed by a small random quantum map, metastable characteristics were observed in the generator spectrum of the initial collision model. Finally, a method of observing slowly decaying metastable behaviour in non-Markovian quantum systems using multi-time correlations is proposed.

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Chapter 1

Introduction

In the study of quantum systems, one of the central technological challenges is the simulation of open quantum systems. The environment that an open quantum systems interacts with can be extremely complex. These interactions transfer information in and out of the system, making it almost impossible to isolate the system whilst preserving its important quantum properties, such as quantum coherence. Despite these difficulties, recently significant advancements have been made in the experimental simulation of open quantum systems [1, 2, 3, 4, 5]. These advancements have allowed for the observation of interesting non-equilibrium phenomena.

One such phenomena we are interested in is Metastability. This phenomenon occurs when a system appears to be in a stationary state which is not the true stationary state of the system [6]. This slowly decaying or stationary behaviour is called the metastable regime. The metastable regime is often the cause of first order phase transitions at a finite system size [7]. Additionally, metastability can also emerge in systems with a complex, but slow relaxation to a stationary state. An interesting contemporary example of this slow, complex relaxation is spin glass [8]. Spin glass, the subject of Giorgio Parisi's Nobel Prize winning research, is a metastable system with fascinating applications in machine learning [9] and the simulation of complex physical systems [10].

However, despite the relevance of metastability in many systems of interest, research into metastability in open quantum systems is still ongoing. We will look at recent results [7, 11, 12] that characterise metastability in open quantum systems by means of a separation in the generator spectrum and establish a minimum separation required to warrant metastability in open quantum systems. Aiming to further explore the mechanisms behind metastability.

A different way to study metastability is via time (or temporal) correlations. These correlations measure how much information is shared between a system at two or more points in time. Beyond the standard two-point correlation, multi-time correlations can be used to study non-Markovaian memory. Quantum systems can be either Markovian or non-Markovian. Markovian systems are memoryless and do not carry information through time, thus are temporally uncorrelated, whereas non-Markovian systems do carry information through time and rely heavily on the history of their temporal evolution [13, 14]. These multi-time correlations can be observed experimentally [15]. In particular, optical techniques such as pump probe spectroscopy have been shown to measure non-Markovian characteristics [16]. Using these correlations, we explore the possibility of utilising memory to observe metastable behaviour in non-Markovian systems[1].

Building upon current research on metastability in open quantum systems, this thesis delves deeper into possible mechanisms behind metastability and the methods used to observe it. In Chapter 2, a theory of metastability in particular, the spectral criteria for metastability is set. Chapter 3, introduces a quantum three-body collision model we call the partial swap model, and investigates the slow decay to a

[1] It is possible to have Markovian metastable systems, but we study a non-Markovian system that is metastable.

steady state. Chapter 4 explores whether metastability manifests in the partial swap model and multiple variations on this model. Lastly, Chapter 5 explores memory in the partial swap models using multi-time correlations and examines possible connections to metastability.

1.1 Preliminaries

Note: This section provides readers with a brief background on the mathematical objects used throughout this thesis to describe quantum systems.

In this thesis, we represent the qubit states of a system using a density matrix or density operator. The density matrix is a mathematically convenient way of representing qubit states and is equivalent to representing the qubits at state vectors [17]. In general, a quantum state can be represented by an ensemble of pure states $|\psi_i\rangle$ with probability p_i. The density matrix of this state is,

$$\rho \equiv \sum_i p_i \left|\psi_i\right\rangle\left\langle\psi_i\right| . \tag{1.1}$$

It is normalised such that $\sum_i p_i = 1$. This normalisation requires all valid density matrices to have a trace of 1. As the outer product of a pure state $|\psi_i\rangle\langle\psi_i|$ is hermitian and the probabilities p_i are all real numbers, the density matrix must also be hermitian, such that $\rho = \rho^\dagger$. To construct a random $n \times n$ density matrix, a random complex matrix $A \in \mathbb{C}^{n\times n}$ is multiplied by the complex conjugate of itself, to ensure hermiticity, and divided by its trace, to ensure normalisation.

$$\rho_A = \frac{AA^\dagger}{Tr[AA^\dagger]} \tag{1.2}$$

The evolution of a density matrix is described by a completely positive, trace preserving (CPTP) or quantum map $\mathcal{E}$. $\mathcal{E}$ maps an input density matrix (ρ) to an output density matrix (ρ') [18].

$$\rho \mapsto \rho^{'} = \mathcal{E}(\rho) \tag{1.3}$$

As this map is completely positive and trace preserving, the output state (ρ') is hermitian and trace 1.

CPTP maps can also be used to derive the Lindblad master equation [19]. The Lindblad master equation is further explained in section 2.1.

A useful representation of a CPTP map called the Choi matrix. The CPTP map $\mathcal{E}$ can be converted to its Choi form $\Upsilon_\mathcal{E}$ by adding an ancilla. For a one qubit map, the input density matrix is replaced with a maximally entangled Bell state. As only one qubit in the Bell state is acted on my the map, an identity matrix is added onto the space of the ancilla [13].

$$\Upsilon_\mathcal{E} = \mathbb{I} \otimes \mathcal{E}(|\Phi^+\rangle\langle\Phi^+|) \tag{1.4}$$

Diagrammatically, this is equivalent to 'bending' one of the legs to the other side.

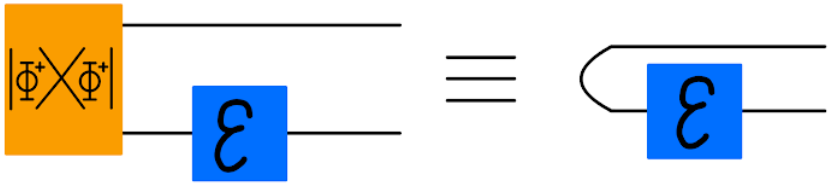


Figure 1.1: The Choi state $\Upsilon_\mathcal{E}$ of CPTP map $\mathcal{E}$

The CPTP map $\mathcal{E}(\rho)$ can be written in terms of its Choi state $\Upsilon_{\mathcal{E}}$. Where $Tr_i[\cdots]$ is the partial trace over the input space.

$$\mathcal{E} = Tr_i[(\rho^T \otimes \mathbb{I})\Upsilon_{\mathcal{E}}] \tag{1.5}$$

Equation 1.5 is represented pictorially by figure 1.2.

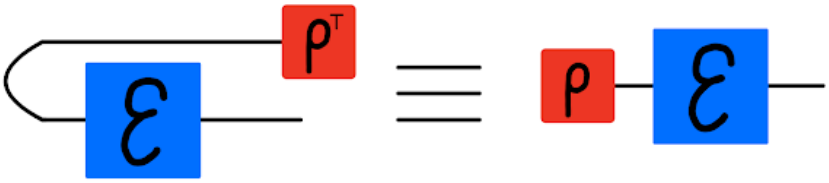


Figure 1.2: The CPTP map $\mathcal{E}(\rho)$ written in terms of its Choi state $\Upsilon_{\mathcal{E}}$

Chapter 2

Metastability in Open Quantum Systems

Metastability is the occurrence of an apparent stationary state for a distinct timescale, after which the system subsequently decays to its true stationary state [7, 6]. It occurs in many everyday systems. For example, classical systems like amorphous solids (e.g. glass) are metastable in their glassy state [20]. Metastability of course, can also occur in quantum systems. In an isolated (closed) quantum system, metastability occurs as a consequence of multiple local minima in the system's free energy [7]. However, for open systems such a general description remains elusive. Although metastability can be the consequence of first order phase transitions [12], we will focus on more general instances where metastability occurs as a result of complex, slow relaxation to a stationary state.

Currently, research into metastable open quantum systems is ongoing. The main criteria used is the spectral theory of metastability [11]. This theory states that all metastable systems must have a large separation in the spectrum of their generator eigenvalues. A large separation refers to a gap of multiple orders of magnitude between two neighbouring eigenvalues (λ_m and λ_{m+1}). The mystery is the exact mechanisms which cause metastability in open quantum systems. That is, how can we engineer a metastable system? Are there certain properties of open quantum systems we can alter to make the system metastable?

2.1 The Master Equation

The dynamics of an open quantum system are governed by the Master Equation (2.1) [22, 23].

$$\frac{d}{dt}\rho_t = \mathcal{L}(\rho_t) \tag{2.1}$$

It is a generalisation of the Schrödinger equation for open quantum systems. The generator $\mathcal{L}$ of the dynamics is

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_j [J_j \rho J_j^\dagger - \frac{1}{2}\{J_j^\dagger J_j, \rho\}] \tag{2.2}$$

where H is the system Hamiltonian and $\{J_j\}$ are the quantum jump operators. The jump operators help describe system environment interactions. Equation 2.1 can be solved in terms of the final state ρ_t. Allowing the dynamics of the system to be described by the dynamical map Λ_t,

$$\rho_t = e^{t\mathcal{L}}(\rho_0) = \Lambda_t(\rho_0). \tag{2.3}$$

Whilst $\mathcal{L}$ generally is not diagonalisable, one can still infer its eigenvalues λ_k from its Jordan decomposition. Each eigenvalue has a corresponding R_k eigenmatrix or Jordan block. As the generator represents the dynamics of a legitimate quantum system, it is completely positive and trace preserving [24]. This results in the first eigenvalue of $\mathcal{L}$ $\lambda_1 = 0$. Assuming relaxation to a unique stationary state, the first eigenmatrix is $R_1 = \rho_{ss}$. The real part of all other eigenvalues are negative, $Re[\lambda_k] \leq 0$. These properties allow for the spectral decomposition of $\mathcal{L}$ [25].

2.1.1 Spectral Decomposition of the Generator

To investigate the spectral properties of the generator $\mathcal{L}$, equation 2.3 is rewritten, transforming the generator to its Jordan Canonical form. In this form, Equation 2.3 can be written in terms of its eigenvalues λ_k, and left (L_k) and right (R_k) eigenmatrices. Note, L_k are the eigenmatrices of $\mathcal{L}^\dagger$ and are normalised such that $\mathrm{Tr}(L_j R_k) = \delta_{jk}$. This normalisation implies $L_1 = \mathbb{1}$[7, 25, 26].

$$\rho(t) = e^{t\mathcal{L}}[\rho(0)] = \sum_{k=1} e^{t\lambda_k} \mathrm{Tr}[L_k \rho(0)] R_k \tag{2.4}$$

Letting $c_k = \mathrm{Tr}[L_k \rho(0)]$, when $k = 1$, c_k is also equal to 1 $c_1 = 1$. Thus when $k = 1$, $c_k e^{t\lambda_k} R_k = \rho_{ss}$. Writing this term outside of the sum,

$$\rho(t) = \rho_{ss} + \sum_{k=2} c_k e^{t\lambda_k} R_k. \tag{2.5}$$

This equation is the basis of the spectral theory of metastability.

2.2 Spectral Theory of Metastability

The spectral theory states that metastability corresponds to a large separation in the real part of the eigenvalue spectrum [12]. When there is a large separation such that $|Re[\lambda_m]| \ll |Re[\lambda_{m+1}]|$ (for some non-zero eigenvalues $m \geq 2$), the non-zero eigenvalues can be split into two groups. One group which decays very quickly ($k \geq m+1$) and another which decays much slower ($2 \leq k \leq m$). Looking at the sum term in equation 2.5, when there exists a large separation in the eigenvalues, there will exist a time regime $t_1 \leq t \leq t_2$ where:

$$c_m e^{t\lambda_m} R_m \gg c_{m+1} e^{t\lambda_{m+1}} R_{m+1} \tag{2.6}$$

During this time, only the slow decaying eigenvalues will make a meaningful contribution to the dynamics of the system, causing the system to appear stationary. After this time regime, as $t \to \infty$ the slow decaying terms decay to 0 and only the stationary state ρ_{ss} survives. Whilst the exact size of the gap needed is ambiguous, the separation must be large enough such that the eigenvalues on either side of the gap are multiple orders of magnitude apart.

$$\frac{Re[\lambda_m]}{Re[\lambda_{m+1}]} \ll 1 \tag{2.7}$$

Using this theory, we present two small quantum systems, one which is metastable and another comprising of a random CPTP map which we would not expect to be metastable.

2.2.1 Example: Minimal Metastable Quantum system

First we consider a minimal open quantum system, described in [11], that is metastable. In this open quantum system, a spin-1/2 qubit is in the presence of a magnetic field $H = -\omega S_z$. Note S_Z is the spin operator along the z axis. The system has uncorrelated-in-time fluctuations which amount to dephasing $J_z = \sqrt{\gamma} S_z$. It is in contact with an infinite temperature bath described by the jump operators $J_\pm = \sqrt{\kappa/2} S_\pm$, where $S_\pm = S_x \pm S_y$. The eigenvalues of this system are $\lambda = 0, \kappa, (\gamma+\kappa)/2 \mp \mathrm{i}\omega$. When

$\kappa/\gamma \ll 1$ there is a large separation between the second and third eigenvalues. [11] reported that the system is metastable for $\kappa/\gamma = 0.005$ and $\omega/(\gamma+\kappa) = O(1)$. Therefore, we can conclude that if our separation ratio 2.7 is less than 10^{-3}, the system can be considered metastable.

2.2.2 Example: Random CPTP map

In contrast to the metastable system above, we now show that the separation of eigenvalues is unlikely to be seen in a random CPTP map for one qubit. For a one qubit map, $d^2 = 4$ random complex matrices are constructed $\{F_k\}_{k=1}^{d^2} \in \mathbb{C}^{2\times 2}$. From these matrices, F_0 is constructed $F_0 = (\sum_k F_k^\dagger F_k)^{-1/2}$. The random CPTP map $\mathcal{R}$ is constructed such that

$$\mathcal{R} \equiv \sum_k (F_k F_0) \otimes (F_k F_0)^* \tag{2.8}$$

Calculating the eigenvalues of a random CPTP map we obtain, $\lambda_{\mathcal{L}}^R = 0, -0.32$ and -0.89. All of these eigenvalues are of the same order of magnitude, indicating that this system is not metastable.

This difference in the metastability observed in the minimal system and a random CPTP map indicates that the spectral gap is not a given in all open quantum systems and that metastability is a unique phenomenon in some system. This allows us to assume that we are unlikely to observe metastability in some random quantum system.

Chapter 3

The Partial Swap Three-body Collision Model

In order to further our investigations into metastability into open quantum systems, we want to explore a simple instance of metastability. For this, we choose the partial swap three-body collision model. We chose this model as there was a consistent power law decay of the qubits observed in [31]. The partial swap model describes collisions between three qubits, one system qubit and two environment qubits. The collisions between these qubits are described by the partial swap operator.

The partial swap operator (3.1) is a two qubit gate. It partially swaps the system qubit with one of the environment qubits. The strength of this interaction is denoted by η with complete swapping of qubits when $\eta = \pi/2$ and no swapping when $\eta = 0$. During each collision, information is transferred from the system qubit to the environment. The partial swap gate is defined as,

$$U_i = Cos(\eta)\mathbb{1} + \mathbf{i}\, Sin(\eta) S_i. \tag{3.1}$$

Here S_i is the swap operator between the system and environment i, such that $S_i |\psi\rangle \otimes |\phi\rangle = |\phi\rangle \otimes |\psi\rangle$ ($|\psi\rangle$ and $|\phi\rangle$ are pure states). The partial swap operator describes a quantum information process known as Homogenization [27, 28]. In this process, the system qubit ρ interacts sequentially with a series of identically prepared environment qubits ξ via the partial swap operator. As the number of environment degrees of freedom becomes very large, the final system state ρ' is infinitesimally close to the initial environment state ξ irrespective of the system's initial state ρ. In contrast, our model consists of only three qubits. This is the smallest nontrivial environment size.

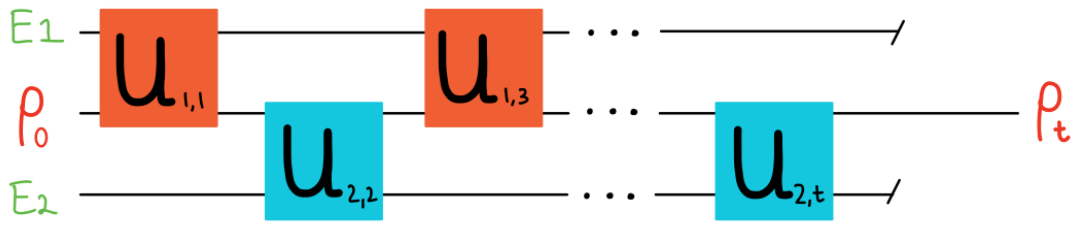


Figure 3.1: A series of partial swap operators acting on the system and either environment one (orange) or environment two (blue). giving the final state at time t ρ_t. ρ' is the density operator of the system after t collisions and i indicates which environment qubit the system interacts with

A two qubit partial swap gate has been experimentally realised using linear optics to partially swap polarisation states of photons [3]. This implementation operates with high fidelity and is easily tunable. The stability of this experimental process was further improved in [4].

3.1 Simulating the Partial Swap Model

In order to simulate the partial swap operation on our three qubit system, the two qubit partial swap gates must be converted to three qubit gates. This is done by acting an identity matrix on the un-swapped qubit. S is the 2 qubit swap operator and $\mathbb{1}_n$ is a $n \times n$ identity matrix. The two partial swap operators for our system are now:

$$U_1 = (Cos(\eta)\mathbb{1}_4 + \mathbf{i}\, Sin(\eta)S) \otimes \mathbb{1}_2 \tag{3.2}$$

$$U_2 = \mathbb{1}_2 \otimes (Cos(\eta)\mathbb{1}_4 + \mathbf{i}\, Sin(\eta)S) \tag{3.3}$$

The initial state of our system is set to be $\rho = \rho_{E1} \otimes \rho_s \otimes \rho_{E2}$. The state ρ is a pure density matrix, such that $\rho = |\psi\rangle\langle\psi|$ [17]. This state evolves by the unitary operator U_i, defined by the quantum map $\mathcal{E}(\rho)$ as,

$$\rho^{'} = \mathcal{E}(\rho) = U_i\, \rho\, U_i^{\dagger}. \tag{3.4}$$

The system qubit can be obtained by performing a partial trace over the environment qubits.

$$\rho_s^{'} = \mathcal{F}(\rho_s) = \mathrm{Tr}_E[U_i\, \rho\, U_i^{\dagger}] \tag{3.5}$$

Although this transformation is easy to construct, it is not mathematically convenient. In this form, the quantum map cannot be represented independently of the initial state. To represent the map independently, both the quantum map and initial state are vectorised as,

$$|\rho^{'}\rangle\rangle = |\mathcal{E}(\rho)\rangle\rangle = \mathcal{E}_A|\rho\rangle\rangle \tag{3.6}$$

With the vectorised state $|\rho\rangle\rangle = \sum_{rs}(\rho)_{rs}\,|rs\rangle$ for $\rho = \sum_{rs}(\rho_{rs}\,|r\rangle\langle s|)$. The vectorised map, as described by Sudarshan *et al.* [29], is an alternate representation of the quantum map $\mathcal{E}$ known as the A-form. In this form, the tensor product of the unitary operator and its conjugate is taken to form the map,

$$\mathcal{U}_i = (U_i \otimes U_i^*). \tag{3.7}$$

To represent our partial swap gates independently, we construct two A-form non-unitary maps, $\mathcal{U}_1$ and $\mathcal{U}_2$ such that,

$$\mathcal{U}_1 = (U_1 \otimes U_1^*) \tag{3.8}$$

and

$$\mathcal{U}_2 = (U_2 \otimes U_2^*) \tag{3.9}$$

Acting these partial swap maps on a vectorised state $|\rho\rangle\rangle$ is equivalent to equation 3.4.

Extending this to multiple timesteps, the A-form maps act on the state $|\rho_0\rangle\rangle$ t times resulting in the state ρ_t. This is representative of our system qubit undergoing a partial swap with the environment qubits t times as,

$$|\rho_t\rangle\rangle = \mathcal{U}_{i,t} \cdots \mathcal{U}_{i,2}\, \mathcal{U}_{i,1}|\rho_0\rangle\rangle. \tag{3.10}$$

The system can undergo ordered swaps (e.g. $\mathcal{U}_1\, \mathcal{U}_2\, \mathcal{U}_1\, \mathcal{U}_2 \cdots$) or the swaps can be random (e.g. $\mathcal{U}_1\, \mathcal{U}_1\, \mathcal{U}_2\, \mathcal{U}_1 \cdots$). Note, there are 2^t possible random swap sequences for a given time t.

3.1.1 System Dynamics

To look at the dynamics of the system through time, we measure the survival probability of the system qubit b_z. Survival probability measures how much of the system 'survives' after a given time. We calculate survival probability of the system using the Pauli z matrix $\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Acting σ_z on the system qubit gives the operator $P_z = (\mathbb{1} \otimes \sigma_z \otimes \mathbb{1})$.

$$b_z(t) = \mathrm{Tr}[(P_z \otimes P_z^*)|\rho_t\rangle\rangle] \tag{3.11}$$

Despite its name, the survival probability does not necessarily decay monotonically over time [30]. It can also temporarily increase or oscillate over time with or without decay. Using Python to calculate

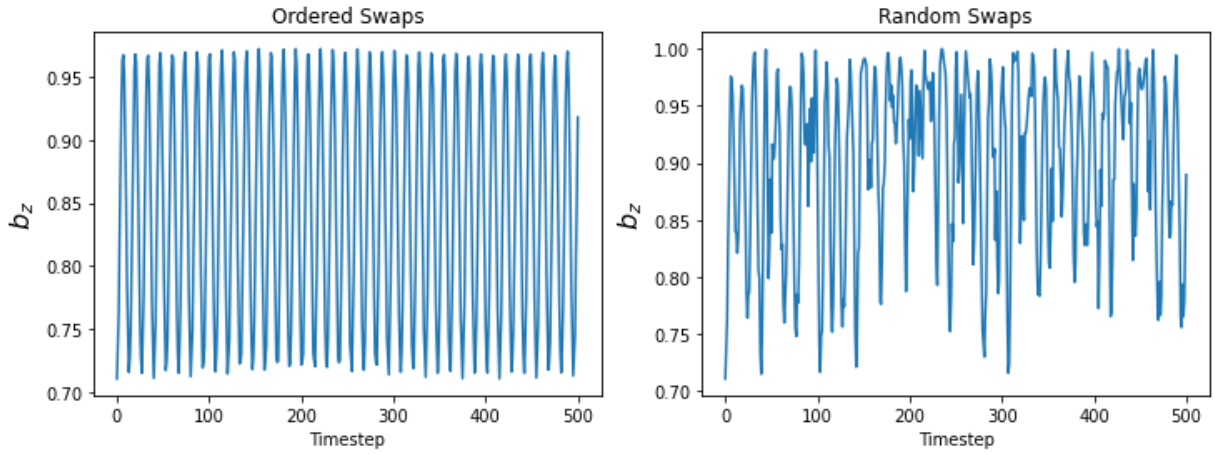


Figure 3.2: The survival probability of our system when acted on by ordered partial swaps (left) and a random sequence of partial swaps (right). The environments are each initialised to the $|0\rangle\langle 0|$ state and our system qubit is a random density matrix and $\eta = \pi/10$

(3.11), we generate figure 3.2 for our partial swap model. We observe a clear difference in the dynamics of the ordered swapped and randomly swapped systems. The oscillations of the ordered swap system are consistent and do not decay over time, whereas the oscillations of a randomly swapped system are not consistent. This could indicate a decay of the system qubit over time. To facilitate easier calculations, we construct an average dynamical map Λ_t. This average map describes the average decay of the system qubit in a randomly swapped partial swap collision model.

3.2 The Partial Swap Average Map

We define our average map as a probabilistic mixture of the partial swap maps $\mathcal{U}_1$ and $\mathcal{U}_2$. To represent a random sequence of swaps, the probability of either swap is 1/2.

$$\mathcal{U}_A = \sum_{i=1}^{2} p_i \mathcal{U}_i = \frac{1}{2}(\mathcal{U}_1 + \mathcal{U}_2) \tag{3.12}$$

Extending this to multiple timesteps, we construct our average dynamical map,

$$\Lambda_t = \mathcal{U}_{A,t} \cdots \mathcal{U}_{A,2} \cdot \mathcal{U}_{A,1} = \mathcal{U}_A^t \tag{3.13}$$

Acting this average map on our initial state is equivalent to the average of all random sequences of swaps acting on our initial state. With an interaction strength of $\eta = \pi/10$, we used Python to plot the average

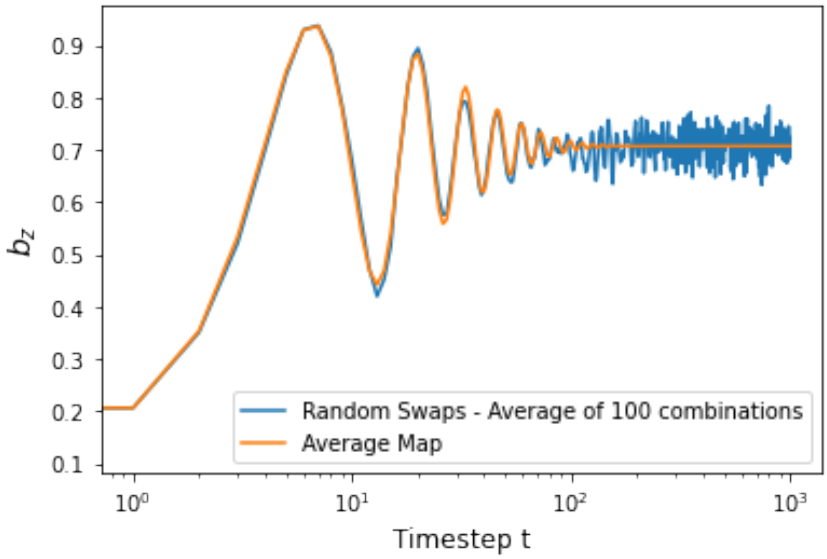


Figure 3.3: The survival probability of the system qubit when acted on by the average dynamical map Λ_t compared to an average of 100 random swap sequences. Errors are all within one standard deviation ($\sigma = 1/10$).

map's survival probability. Figure 3.3 shows this against an average of 100 different random sequences. The average map seems to accurately represent the dynamics of the random swap system. Looking at the survival probability of our system for different η (figure 3.4), we see an increase in the decay time as η decreases. Each system reaches a steady state around time $1/\eta^2$. This means that systems with a large interaction strength, e.g. $\eta = \pi/10$, decay quickly.

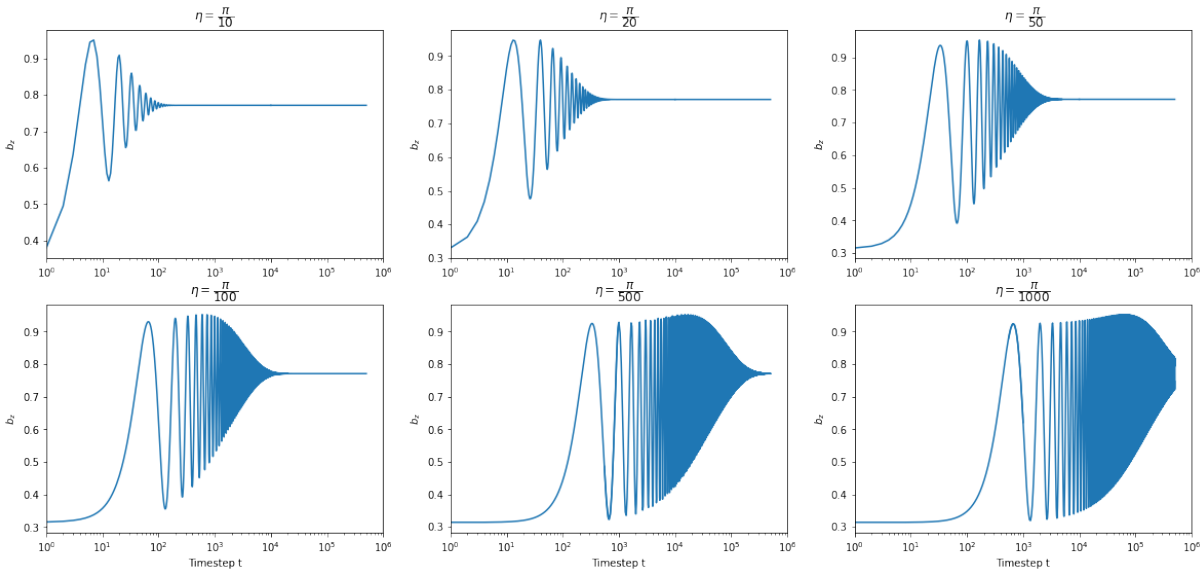


Figure 3.4: The survival probability of the system qubit for different interaction strengths η

Taking a closer look at the peaks of the survival probability in figure 3.5 when $\eta = \pi/1000$. We observe a straight line between $t = 10^4$ and $t = 10^5$. As this plot is on a log scale, this straight line shows a slow power law decay in the system. We expect this slow decay to indicate metastability in the system.

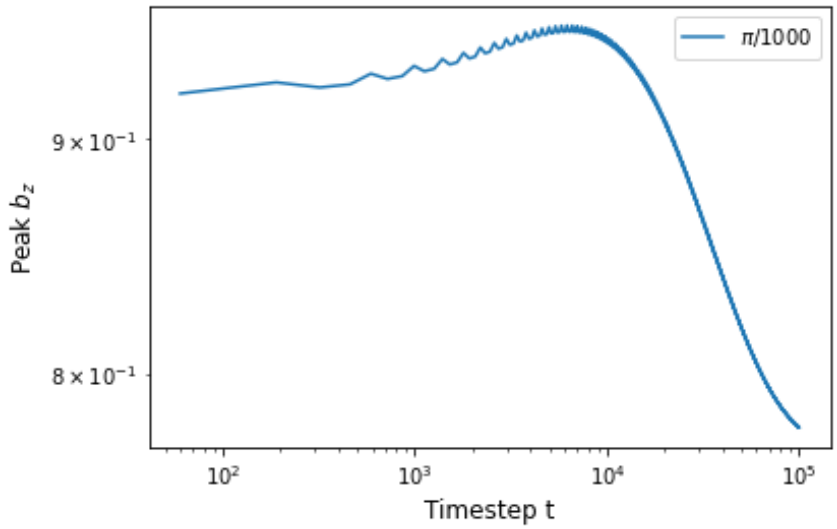


Figure 3.5: Peaks of the survival probability of the partial swap system for $\eta = \pi/1000$. A power law decay is observe around time $t = 10^4 tot = 10^5$

3.2.1 Results of Benenti *et al.*

Benenti *et al.*[31] showed a power law decay of qubits when acted on by a random sequence of partial swap operators with interaction strength $\eta = \pi/10$, shown in figure 3.6. This power law decay suggested metastability. However, the value measured is a time average of the survival probability b_z. Here $\mathcal{B}$ is the sum of the survival probability for each qubit (System, Environment 1 and Environment 2) at $t = 0$.

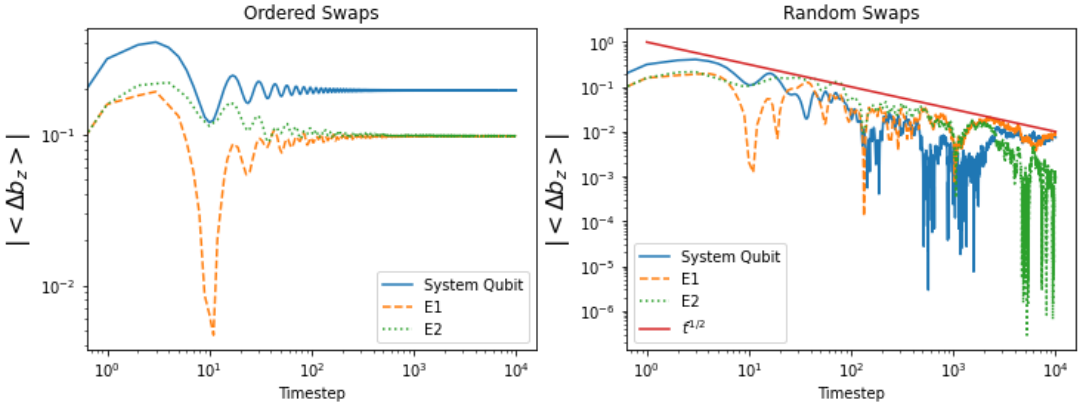


Figure 3.6: Reproduction of results. Time averaged survival probability of the system qubit for both ordered and random swaps with interaction strength $\eta = \pi/10$.

$$|< \Delta b_z >|(T) = \frac{1}{T+1} \sum_{t=0}^{T} b_z(t) - \frac{\mathcal{B}}{3} \tag{3.14}$$

However, the powerlaw decay observed is not an accurate description of the dynamics of the system. On closer inspection of equation 3.14, if the system survival probability reaches a stationary state of $\mathcal{B}/3$ at time s, for $T > s$ the sum will be a constant. This causes the resulting time average $|< \Delta b_z >|$ to be dominated by the term $1/T$ for large time T. This term is the cause of the power law decay in figure 3.6.

Chapter 4

Metastability in Partial Swap Models

To verify if the partial swap system is metastable, we use the spectral theory of metastability. From the average dynamical map Λ_t in 3.13, we calculate the 64 generator eigenvalues of the partial swap system with interaction strength $\eta = \pi/1000$. Using equation 2.3, we replace the generator and dynamical map with their respective eigenvalues and take the Taylor series expansion of the exponential term to approximate $\lambda_{\mathcal{L}}$.

$$\lambda_{\Lambda_t} = e^{t\lambda_{\mathcal{L}}} = 1 + t\lambda_{\mathcal{L}} + \cdots \tag{4.1}$$

$$\lambda_{\mathcal{L}} \approx \frac{\lambda_{\Lambda_t} - 1}{t} \tag{4.2}$$

We observe large degeneracies in the eigenvalues (These degeneracies arise because of symmetries in the map). Out of the 64 eigenvalues there are 8 unique values and only 4 unique real parts. The unique eigenvalues are:

$$Re[\lambda_{\mathcal{L}}] = \{0, -4.9348 \times 10^{-6}, -1.2337 \times 10^{-5}, -1.4804 \times 10^{-5}\}. \tag{4.3}$$

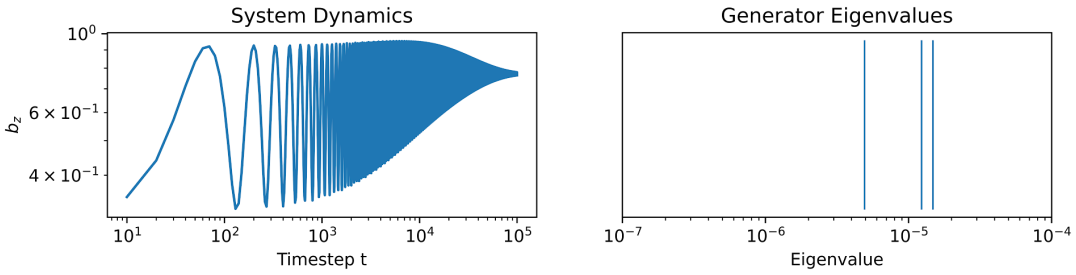


Figure 4.1: Model 0: survival probability (left) and spectrum (right) of non-zero generator eigenvalues $|Re[\lambda_{\mathcal{L}}]|$.

These eigenvalues are all around the same order of magnitude (10^{-5}). The largest separation is between $\lambda_m = -4.9348 \times 10^{-6}$ and $\lambda_{m+1} = -1.2337 \times 10^{-5}$. Here the ratio $\lambda_m/\lambda_{m+1} = 0.4$. This ratio is not small. Therefore the partial swap system is not metastable by the spectral theory of metastability. This contradicts the slow power law decay observed in figure 3.5 which we expected to indicate metastability. One possibility for this is that the system is not metastable and the power law decay observed is not slow enough to cause metastability in the system. Another possibility is that the large degeneracies observed in the eigenvalue spectrum hide the separation. Is there a way that we could break the degeneracies so that the eigenvalues show a large separation?

4.1 Modifying the Partial Swap Model

To investigate the dynamics a metastable system can exhibit, we construct 6 new partial swap models with similar system decay to the partial swap model introduced in Chapter 3. We introduce two new operators to do this. The environment swap operator S_E and the environment partial swap operator U_3. These operators represent interactions between the environment qubits and do not cause interactions with the system qubit. The environment swap operator S_E fully swaps the two environment qubits and the environment partial swap operator U_3 partially swaps the two environment qubits with interaction strength η. We convert these operators to A-form maps $\mathcal{S}_E$ and $\mathcal{U}_3$.

$$S_E = (S \otimes \mathbb{1}) \cdot (\mathbb{1} \otimes S) \cdot (S \otimes \mathbb{1}) \tag{4.4}$$

$$U_3 = (\mathbb{1} \otimes S) \cdot U_1 \cdot (\mathbb{1} \otimes S) \tag{4.5}$$

Using these maps, we create the new partial swap models. Model 0 is the original partial swap system in equation 3.12.

4.1.1 Model 1

This model represents a random choice of either partial swap gate $\mathcal{U}_1$, $\mathcal{U}_2$ or the environment swap gate $\mathcal{S}_E$. Similar to model 0, we create an average map with equal probability of $\mathcal{U}_1$, $\mathcal{U}_2$ or $\mathcal{S}_E$ occurring. We use a small interaction strength $\eta = \pi/1000$ for the partial swaps.

$$\mathcal{E}_1(\eta) = \frac{\mathcal{U}_1(\eta) + \mathcal{U}_2(\eta) + \mathcal{S}_E}{3} \tag{4.6}$$

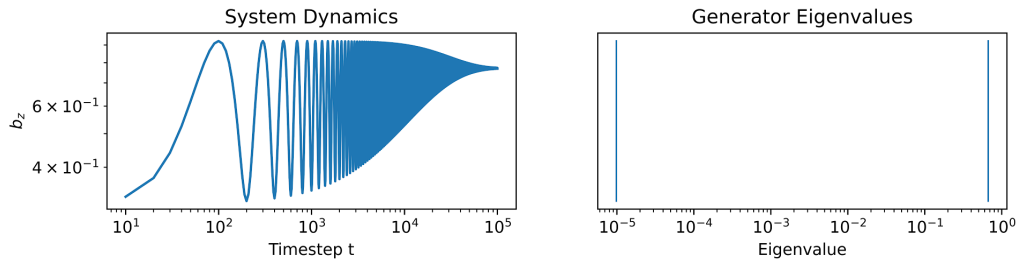


Figure 4.2: Model 1: survival probability (left) and spectrum (right) of non-zero generator eigenvalues $|Re[\lambda_{\mathcal{L}}]|$.

The eigenvalues are:

$$Re[\lambda_{\mathcal{L}}] = \{0, -9.86960 \times 10^{-6}, -6.66669 \times 10^{-1}, -6.66674 \times 10^{-1}\}. \tag{4.7}$$

In this system, smallest non-zero eigenvalue is approximately equal to the square of the interaction strength η^2. In this case, $\eta^2 = (\pi/1000)^2 = 9.86960 \times 10^{-6}$. The largest gap is between $\lambda_m = -9.86960 \times 10^{-6}$ and $\lambda_{m+1} = -6.66669 \times 10^{-1}$, the ratio of this gap $\lambda_m/\lambda_{m+1} \approx 1.5 \times 10^{-5}$. As this ratio is smaller than 10^{-3}, the gap is large. Therefore, Model 1 is a metastable system.

4.1.2 Model 2

Model 2 represents a random choice of either partial swap gate $\mathcal{U}_1$ or $\mathcal{U}_2$ which is then acted on by a swap gate $\mathcal{S}_E$ on the environments. This model is equivalent to acting an environment swap gate $\mathcal{S}_E$ after each average map in model 0. We use a small interaction strength $\eta = \pi/1000$ for the partial swap gates.

$$\mathcal{E}_2(\eta_1) = \frac{\mathcal{U}_1(\eta_1) + \mathcal{U}_2(\eta_1)}{2} \cdot \mathcal{S}_E \tag{4.8}$$

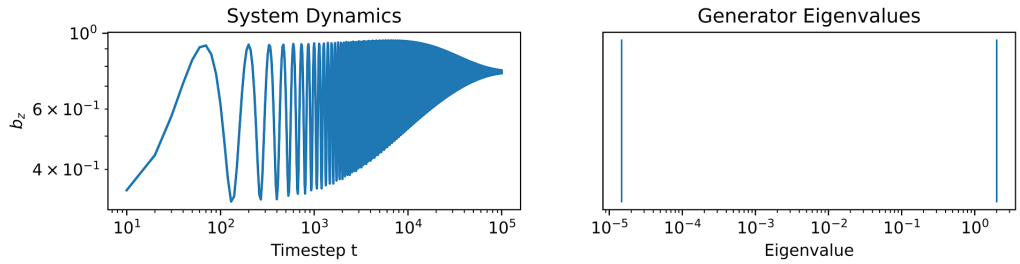


Figure 4.3: Model 2: survival probability (left) and spectrum (right) of non-zero generator eigenvalues $|Re[\lambda_{\mathcal{L}}]|$.

The eigenvalues are:

$$Re[\lambda_{\mathcal{L}}] = \{0, -1.48044 \times 10^{-5}, -1.999987, -1.99999\}. \tag{4.9}$$

The largest gap is between $\lambda_m = -1.48044 \times 10^{-5}$ and $\lambda_{m+1} = -1.999987$. The ratio of this gap $\lambda_m/\lambda_{m+1} \approx 7.4 \times 10^{-6}$. This ratio is very small. Therefore, model 2 is metastable.

4.1.3 Model 3

This model represents a random choice of $\mathcal{U}_1$, $\mathcal{U}_2$ or the environment partial swap $\mathcal{U}_3$ at each time step. This model is similar to model 1 but instead of a full swap of the environments, we use a partial swap with the same interaction strength as the system partial swaps. We use a small interaction strength $\eta = \pi/1000$.

$$\mathcal{E}_3(\eta_1) = \frac{\mathcal{U}_1(\eta_1) + \mathcal{U}_2(\eta_1) + \mathcal{U}_3(\eta_1)}{3} \tag{4.10}$$

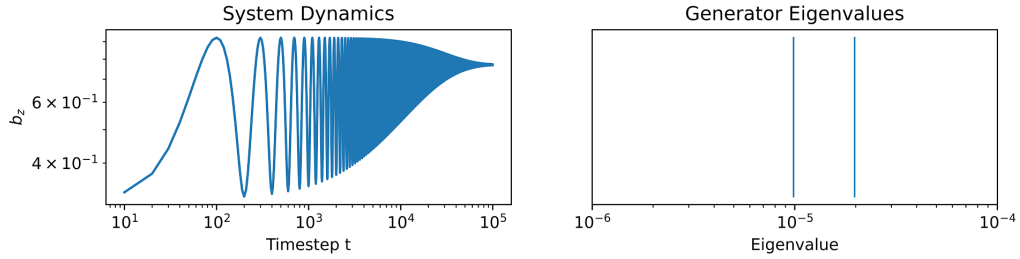


Figure 4.4: Model 3: survival probability (left) and spectrum (right) of non-zero generator eigenvalues $|Re[\lambda_{\mathcal{L}}]|$.

The eigenvalues are:

$$Re[\lambda_{\mathcal{L}}] = \{0, -9.86960 \times 10^{-6}, -1.97391 \times 10^{-5}\}. \tag{4.11}$$

Both non zero eigenvalues are very small. Similar to model 1, the smallest non-zero eigenvalue is approximately equal to η^2. However, in this model the next smallest eigenvalue is still quite small. The gap between the eigenvalues $\lambda_m = -9.86960 \times 10^{-6}$ and $\lambda_{m+1} = -1.97391 \times 10^{-5}$ is quite small, with $\lambda_m/\lambda_{m+1} = 0.5$. Model 3 does not appear to be metastable.

4.1.4 Model 4

This model is similar to model 3, but we instead pick a larger interaction strength for the environment partial swap $\mathcal{U}_3$. We choose the small interaction strength $\eta = \pi/1000$ and the large interaction strength $\eta_1 = \pi/10$.

$$\mathcal{E}_4(\eta_1, \eta_2) = \frac{\mathcal{U}_1(\eta_1) + \mathcal{U}_2(\eta_1) + \mathcal{U}_3(\eta_2)}{3} \tag{4.12}$$

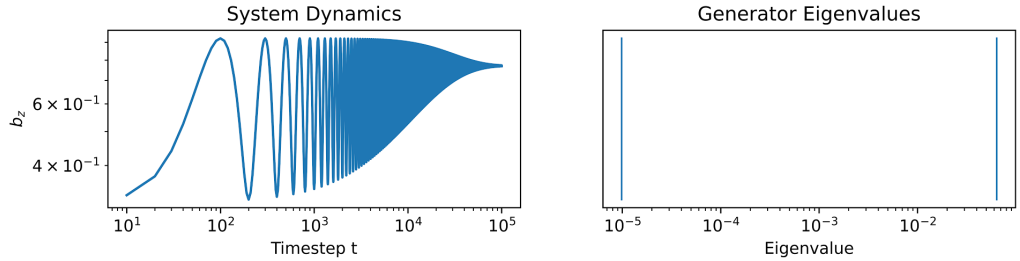


Figure 4.5: Model 4: survival probability (left) and spectrum (right) of non-zero generator eigenvalues $|Re[\lambda_{\mathcal{L}}]|$.

The eigenvalues are:

$$Re[\lambda_{\mathcal{L}}] = \{0, -9.86960 \times 10^{-6}, -6.36643 \times 10^{-2}, -6.36692 \times 10^{-2}\}. \tag{4.13}$$

Similar to model 3 we see the smallest eigenvalue equal to η^2 but now, the next smallest eigenvalue is much larger. The gap between these eigenvalues $\lambda_m = -9.86960 \times 10^{-6}$ and $\lambda_{m+1} = -6.36643 \times 10^{-2}$ is big. Although not as large as other gaps, the ratio of this gap $\lambda_m/\lambda_{m+1} \approx 1.6 \times 10^{-4}$, is still quite small. Therefore, model 4 is metastable.

4.1.5 Model 5

This model is similar to model 2, but instead of acting a full environment swap after the average map, this model acts the environment partial $\mathcal{U}_3$ afterwards. We pick a small interaction strength $\eta_1 = \pi/1000$ for the system partial swaps and a larger interaction strength for the environment partial swaps $\eta_2 = \pi/10$.

$$\mathcal{E}_5(\eta_1, \eta_2) = \frac{\mathcal{U}_1(\eta_1) + \mathcal{U}_2(\eta_1)}{2} \cdot \mathcal{U}_3(\eta_2) \tag{4.14}$$

The eigenvalues are:

$$Re[\lambda_{\mathcal{L}}] = \{0, -1.48044 \times 10^{-5}, -1.89146 \times 10^{-1}, -1.91910 \times 10^{-1}\}. \tag{4.15}$$

The largest gap is between eigenvalues $\lambda_m = -1.48044 \times 10^{-5}$ and $\lambda_{m+1} = -1.89146 \times 10^{-1}$. The ratio between these eigenvalues $\lambda_m/\lambda_{m+1} \approx 7.8 \times 10^{-5}$ is small. Therefore, model 5 is metastable.

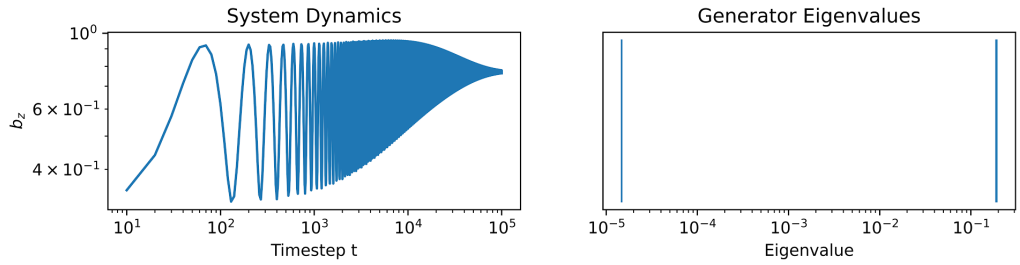


Figure 4.6: Model 5: survival probability (left) and spectrum (right) of non-zero generator eigenvalues $|Re[\lambda_{\mathcal{L}}]|$.

4.1.6 Model 6

This model represents a random choice of any partial swap gate which is then acted on by the environment swap $\mathcal{S}_E$. This model is similar to model 3, but the environment swap map $\mathcal{S}_E$ swaps the environment qubits after each partial swap.

$$\mathcal{E}_6(\eta_1) = \frac{\mathcal{U}_1(\eta_1) + \mathcal{U}_2(\eta_1) + \mathcal{U}_3(\eta_1)}{3} \cdot \mathcal{S}_E \tag{4.16}$$

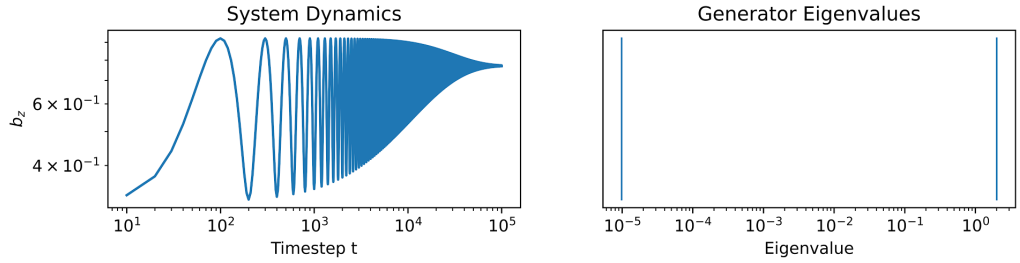


Figure 4.7: Model 6: survival probability (left) and spectrum (right) of non-zero generator eigenvalues $|Re[\lambda_{\mathcal{L}}]|$.

The eigenvalues are:

$$Re[\lambda_{\mathcal{L}}] = \{0, -9.86960 \times 10^{-6}, -1.99998, -1.99999\}. \tag{4.17}$$

These eigenvalues are quite different to the eigenvalues in model 3. Now that we have added the environment swap after each average map we see a separation of the eigenvalues between $\lambda_m = -9.86960 \times 10^{-6}$ and $\lambda_{m+1} = -1.99998$. The ratio of this separation $\lambda_m/\lambda_{m+1} \approx 4.9 \times 10^{-6}$ is small. Therefore model 6 is metastable.

4.2 Dynamics of the Modified Partial Swap Models

As in Chapter 3, we investigate the system dynamics using survival probability. Plotting the survival probabilities of our new models against model 0, we see a pattern. Models 2 and 5 look the same as model 0. This is expected as the operators we added only affect the environment qubits. Acting one of our new operators on the environment after each average map should not change the decay rate of the system qubit. What is interesting is that models 2 and 5 show metastability in their eigenvalue spectrum, but model 0 does not. Since the dynamics appear the same for all 3 models, we would expect that they would either be all metastable or all non-metastable. The other 4 models also appear alike. Models 1, 3, 4 and 6 all have probability 1/3 of each $\mathcal{U}_1$ and $\mathcal{U}_2$. This is what causes the decay rate to look the

same. The metastability of these models is not consistent either. Model 3 did not show metastability in its eigenvalues but models 1, 4 and 6 did.

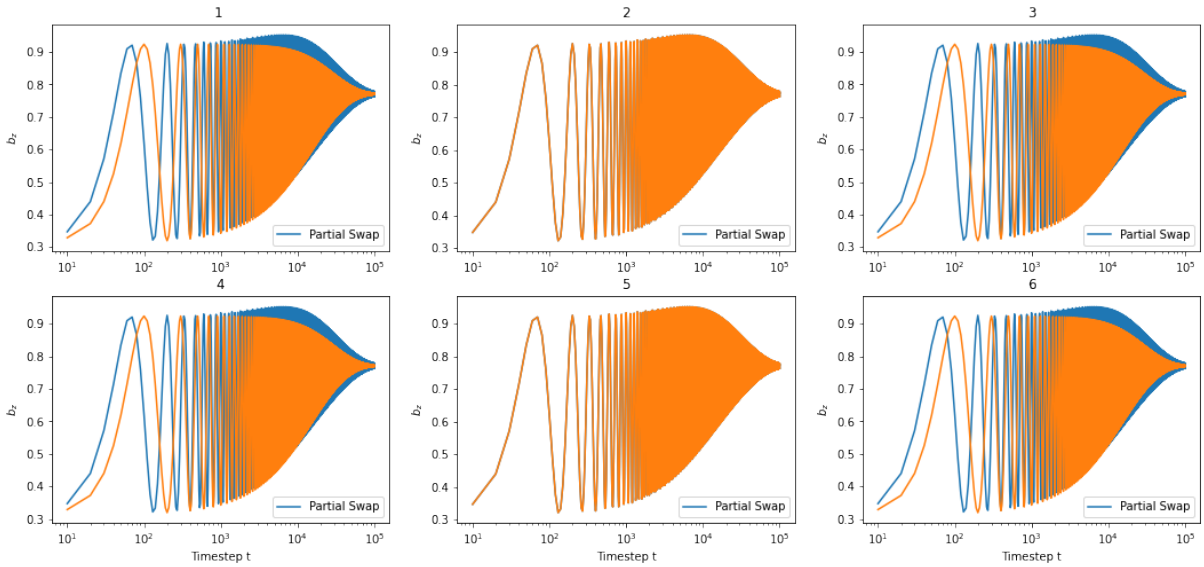


Figure 4.8: The survival probability of models 1-6 against the survival probability of the original partial swap model (model 0). Models 2 and 5 all appear follow the same dynamics as model 0. As model 5 is metastable, this could indicate metastability in model 0 and 2.

Figure 4.9 shows the peaks of the survival probability for models 4 and 5 (represents the two dynamics patterns observed). Between times 10^4 and 10^5 both systems show a power law decay. We approximated this decay with the line for model 4,

$$b_z = 4t^{-1/40} - 2.255 \tag{4.18}$$

and for model 5

$$b_z = 8t^{-1/65} - 5.995. \tag{4.19}$$

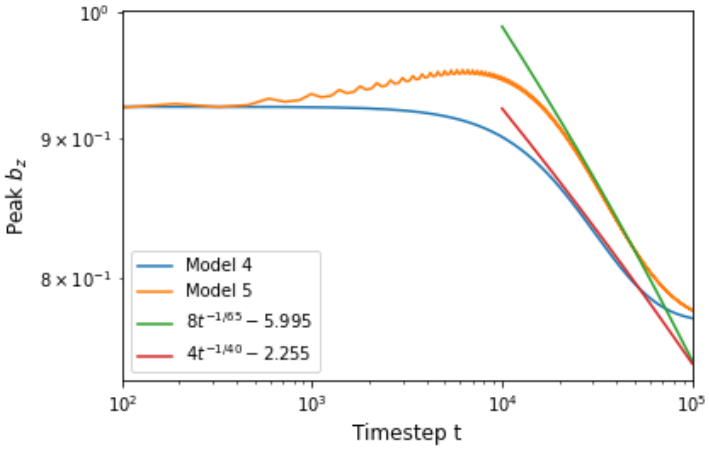


Figure 4.9: Peaks of figure 4.8. Models 0,2 and 5 follow the brown curve and models 1,3,4 and 6 follow the pink. All models have an approximate power law decay between $t = 10^4$ and $t = 10^5$.

This power law decay agrees with the slow decay seen in metastable systems. Although all systems show metastable decay in their survival probability, models 0 and 3 do not have a spectral gap in their generator eigenvalues. Additionally, all models that did show a large separation in their generator eigenvalues had significant swapping between the environment qubits, by significant swapping we refer to a

full swap S_E or a partial swap with a larger interaction strength than the system partial swaps. Why do these large environmental interactions seem to expose metastability in the system? To try and reveal the metastability in the spectrum of model 0 we add a small perturbation to its average map.

4.3 Exposing Metastability in the Partial Swap Model

We added a small perturbation to the average map Λ_t of the partial swap system to try and break the degeneracies [32]. This is done by creating a perturbed map $\mathcal{G}$ using our average map Λ_t and some random map. From equation 2.8, we construct a $d = 8$ random CPTP map $\mathcal{R}$ and create a probabilistic mixture of Λ_t and $\mathcal{R}$.

$$\mathcal{G} = (1 - c)\Lambda_t + c\,\mathcal{R} \tag{4.20}$$

Using a small c, we add a small perturbation to the average map. Looking at the generator eigenvalues of $\mathcal{G}$, we can now see a gap in the spectrum. This separation occurs when $c = 10^{-8}$ and the interaction strength $\eta = \pi/100$.

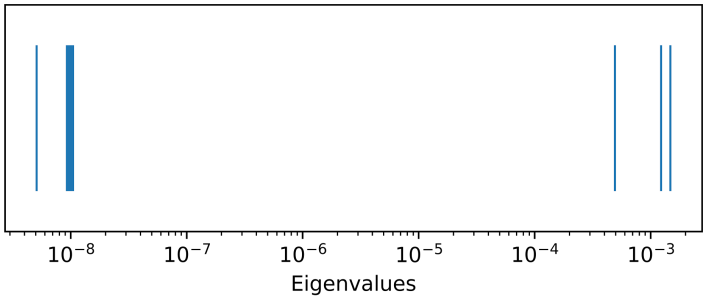


Figure 4.10: Spectrum of the non-zero generator eigenvalues $|Re[\lambda_{\mathcal{L}}]|$. Note that all generator eigenvalues are negative, so $|Re[\lambda_{\mathcal{L}}]| = -Re[\lambda_{\mathcal{L}}]$.

This separation is between eigenvalues $\lambda_m = -1.0500 \times 10^{-8}$ and $\lambda_{m+1} = -4.9333 \times 10^{-4}$. The ratio of this separation is $\lambda_m/\lambda_{m+1} \approx 2.1 \times 10^{-4}$. This ratio is small, so this system can now be considered metastable.

Model 0 shows the same survival probability as metastable systems and when perturbed by a small amount, its generator eigenvalue spectrum shows metastability. This indicates that model 0 is likely a metastable system, but that the metastability is hidden by large degeneracies in its spectrum. Currently, we do not know exactly why these large degeneracies hide the metastable separation or why adding a small perturbation seems to reveal metastable separation in the spectrum. This could be related to the generator eigenvalue separation only appearing in models with large swapping between the environments, but further research is needed to confirm this.

Chapter 5

Memory in Partial Swap Models

In this chapter, we investigate the impact that metastability has on memory and whether we can observe metastability in the time correlations of non-Markoivan open quantum systems.

5.1 Time correlations

Time or temporal correlations measure how much information is carried between different times in a system[33]. Time correlations are a good indicator of the memory in a system. If a system has no correlation throughout time, it must be Markovian or memoryless. If a system does have some correlation then it is non-Markovian.

5.1.1 Three-time correlations

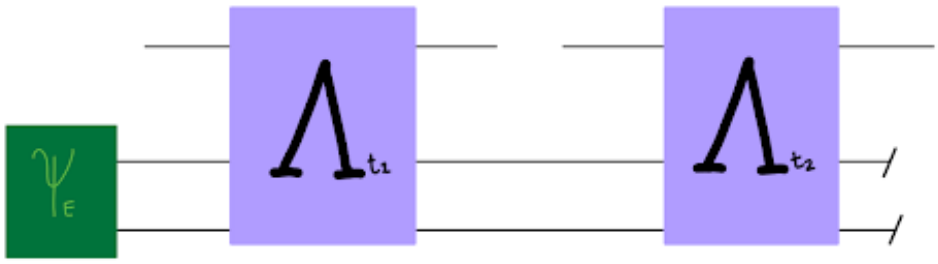


Figure 5.1: Two point time correlation using the dynamical map. Outputs the system qubit state. Note that the qubits are ordered System(Top), Environment 1(Middle), Environment 2(Bottom). To implement this multiplication, we used a python function called NCON (see Appendix A for details)[34].

We measure the two point time correlations of our system between times t_1 and t_2 by measuring the mutual information given by

$$I_\Upsilon(t_1 : t_2) = S(\Upsilon_{t_1}) + S(\Upsilon_{t_2}) - S(\Upsilon_{t_1\, t_2}), \tag{5.1}$$

where $S(\Upsilon) = \sum_i \lambda_i log(\lambda_i)$ is the Von Neumann entropy of the Choi state Υ. This expression is depicted graphically in figure 5.1. The difference between these times is denoted $\Delta t = t_2 - t_1$. Quantum mutual

information is a standard measurement of memory in quantum systems. It is the difference between the Von Neumann entropy of the entire system $\Upsilon_{t_1\,t_2}$ and the system at states Υ_{t_1} and Υ_{t_2} separately. The separate systems Υ_{t_1} and Υ_{t_2} are obtained by tracing out either t_1 or t_2 from the full system $\Upsilon_{t_1\,t_2}$.

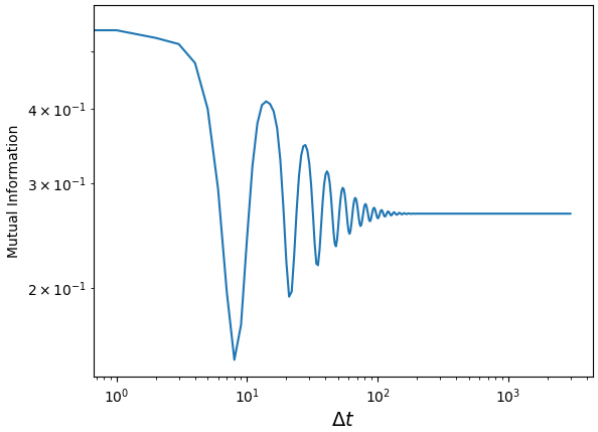


Figure 5.2: Mutual information of the partial swap system (model 0) for $t_1 = 5$ and $\eta = \pi/10$.

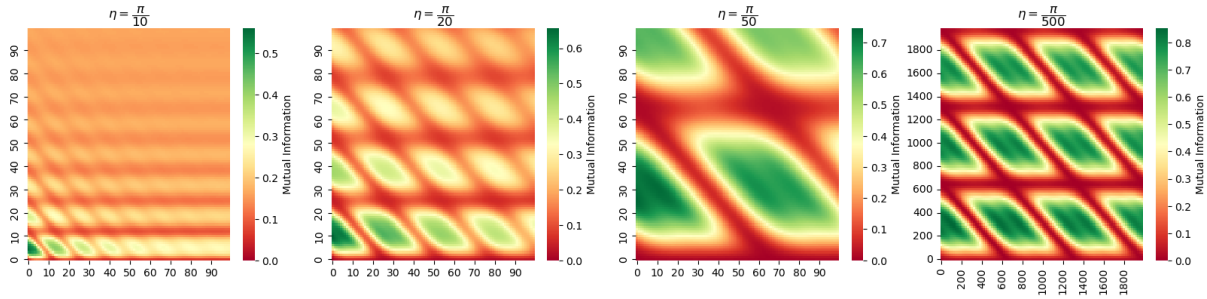


Figure 5.3: Mutual information of the partial swap system (model 0) as Δt(x-axis) and t_1(y-axis) change. As η decreases the pattern in the mutual information decays more slowly.

Calculating the mutual information of the partial swap system (model 0), we observe a power law decay to a steady state. As the mutual information of the partial swap system is non zero. This indicates that the system has memory and is non-Markovian. As η decreases, the decay time also increases. The steady state for all systems appears to occur around η^{-2}. This slow decay of systems with a small η indicates metastability in the system and is consistent with the survival probability of these systems.

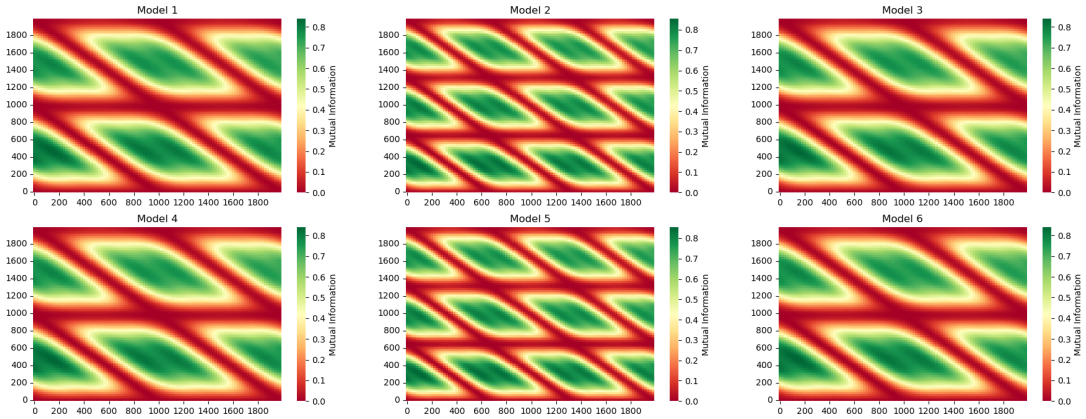


Figure 5.4: Mutual information of models 1-6 as Δt(x-axis) and t_1(y-axis) change. Here small $\eta = \pi/500$ and large $\eta = \pi/5$.

Comparing the mutual information between different partial swap models (figure 5.4), all models are have memory. Similar to the patterns observed in the survival probability, models 0, 2 and 5 follow the one pattern and models 1, 3, 4 and 6 follow another. The main difference in the observed patterns is the amount of peaks. Models 0, 2 and 5 have a shorter period between peaks. This is likely the result of the higher probability of system partial swaps ($\mathcal{U}_1$ and $\mathcal{U}_2$) in these systems. This higher probability means that models 0, 2 and 5 contain more interactions with the system qubit per time step.

5.1.2 Four-time correlations

From the heat maps above we can calculate a four point time correlation. This correlation is calculated from the mutual information $I_\Upsilon(t_2 t_+ : t_- t_1)$ along some line on the heat maps above given by,

$$t_1 = \alpha \Delta t. \tag{5.2}$$

This line when $\alpha = 1$ is plotted in figure 5.5.

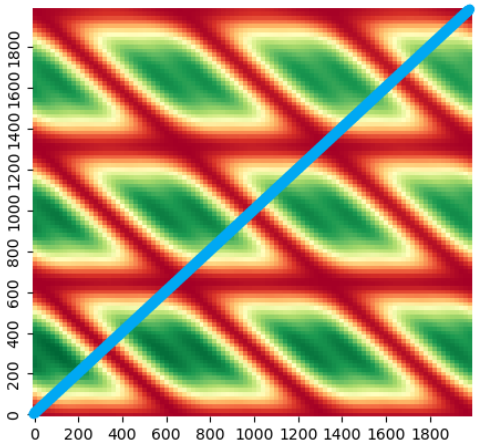
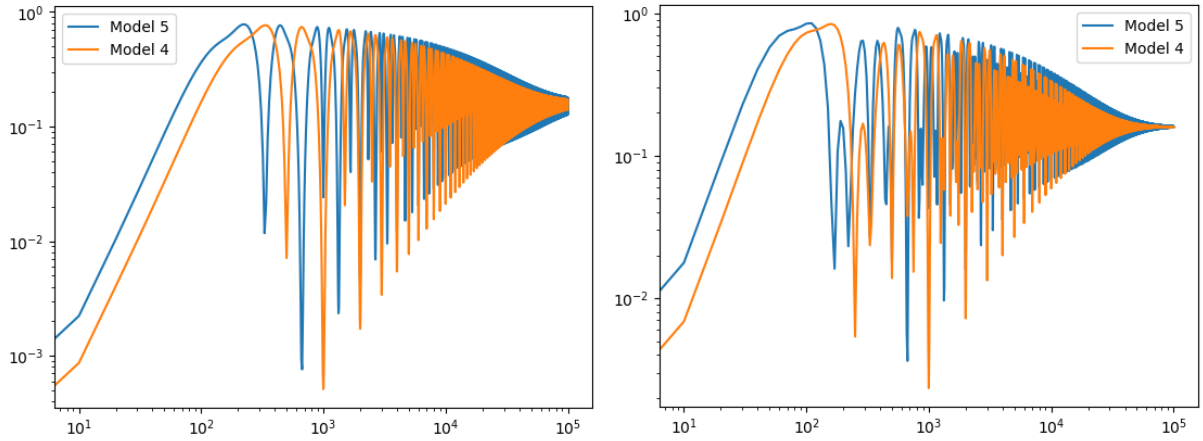


Figure 5.5: Heat map of 2 point time correlations for model 0 and $\eta = \pi/500$ with the straight line $t_1 = \Delta t$(blue).



(a) $\alpha = 1$ (b) $\alpha = 3$

Figure 5.6: 4 point time correlations for Models 4 and 5 along the line (a)$t_1 = \Delta t$ and (b)$t_1 = 3\Delta t$. With small $\eta = \pi/500$ and large $\eta = \pi/5$. Both of these show power law decay at large t

Now plotting the mutual information, figure 5.6 shows a slow power law decay to the same steady

state for both models. Similar to the decay of survival probability, model 5 decays slower than model 4. This power law decay suggests metastability in all the models. With $\alpha = 3$, we still see slow decay, but in this model the decay is slightly faster.

The observation of power law decay in both time correlations is consistent with the survival probability observations. The slow decay of multiple observables of this system is consistent with metastability. Therefore, we can detect metastability in non-Markovian systems by observing slow decay patterns in both the survival probability and time correlations. In particular, as four point time correlations can be measured experimentally using pump probe spectroscopy [16], we have a nice way to detect metastability in non-Markovian quantum systems.

Conclusion

Using the spectral theory of metastability, we investigated the metastability of a simple three-body collision model called the partial swap model. Although the generator spectrum of this model did not show metastability initially, we created variations of this model that did show metastability. These variations only contained additional interactions between the environment qubits, such that the appearance of the dynamics of the system were unaffected (shown in the similarities of both survival probability and time correlation). This difference in spectral criteria exhibited in systems with very similar dynamics is unexpected. As a result of this, we investigated metastability in the original partial swap model further. We found that when perturbing the dynamical map (3.13) by a small amount, a metastable spectrum gap was exposed. This suggests that the partial swap model is in fact metastable for a small η. The reason for this 'hidden' metastability could be the large degeneracies exhibited in the initial eigenvalue spectrum, but further research is needed to confirm this. Additionally, all of the variations that showed spectral metastability contained significant interactions between the environments. The reasoning for this is also unknown.

Finally, we proposed a method of observing metastability in non-Markovian quantum systems. This method uses four-time correlations to observe slow, metastable decay in the system. This method is of particular interest, as four-time correlations can be measured experimentally. Opening up the possibility of experimentally observing metastability in complex non-Markovian quantum systems.

We constructed this simple model to serve as a foundation of studying metastable phenomenon in open quantum systems. Furthering our research of this simple system, we hope to advance our understanding of the mechanisms behind metastability and eventually engineer a metastable open quantum system.

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Appendix A

NCON Integration

NCON or 'Network CONtracter [34] is a python function that we used to compute complex matrix multiplication, tensor products and partial traces. Using this function, we can easily multiply together a complex network of interconnected multi-dimensional arrays or tensors, For example, we used it to compute the multiplications in figure 5.1. To illustrate how NCON works we begin with a simple example of matrix multiplication from [34] $C = AB$. Using ncon() the matrix C can be constructed from a list of tensors $[A, B]$ as C = ncon(tensorList, legList). Where legList = [[-1, 1],[1,-2]] is a python list of the index labels assigned for matrices A and B. Open or uncontracted indices in a multiplication are assigned negative integers whereas the indices that are contracted or summed over are assigned positive integers, as illustrated in Figure A.1. The output tensor indices are ordered $[-1, -2, -3, \cdots]$. The red numbers inside the tensors A, B and C represent the index

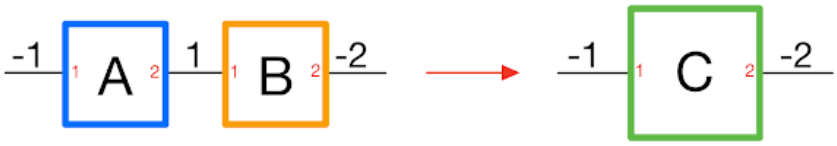


Figure A.1: NCON multiplication: $C = ncon([A, B], [[-1, 1], [1, -2]])$

order of each tensor. This is the order in which the index of the legs should appear in for each tensor in legList.

The ncon call for a simple tensor product $A \otimes B$ is shown in figure A.4. In this call only negative leg numbers are used, as there are no indexes being multiplied. This creates one new object $(A \otimes B)$ with $2 + 2 = 4$ legs. A simple partial trace $\mathrm{Tr}[A \otimes B]$ is shown in figure A.3. In this ncon call, the legs of the two traced out indices are both labelled 1, this sums over the legs, so they are not in the output object.

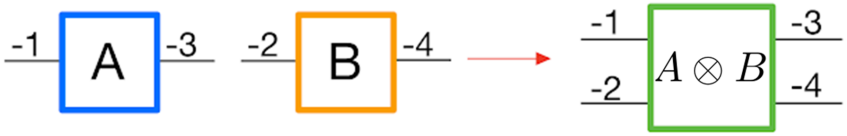


Figure A.2: NCON tensor product: $A \otimes B = ncon([A, B], [[-1, -3], [-2, -4]])$



Figure A.3: NCON Trace: $\mathrm{Tr}_B[A \otimes B] = \mathrm{ncon}([A \otimes B], [[-1, 1, -2, 1])$

A.1 NCON for the Partial Swap Collision Model

When simulating the partial swap system, the maps are all in the A-form. In the A-form, the quantum maps are a tensor product of the operator with their conjugate ($\mathcal{U}_i = U_i \otimes U_i^*$). We construct the quantum map $\mathcal{U}_i$ and Choi state $\Upsilon_{\mathcal{U}_i}$ as shown in figure A.4.

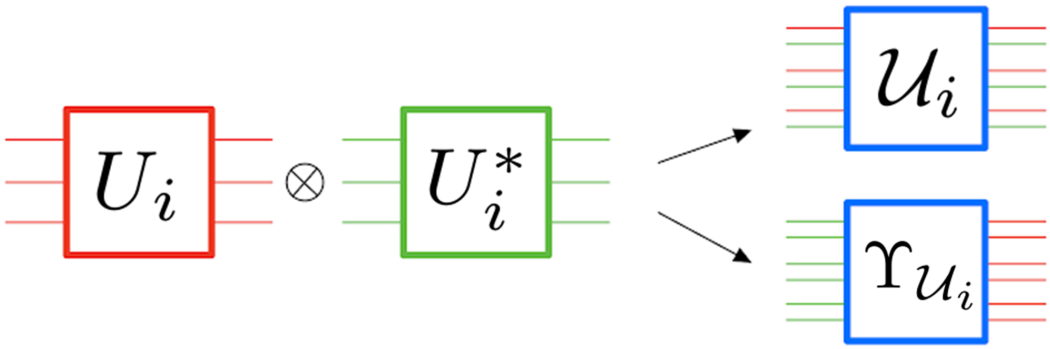


Figure A.4: Construction of quantum map (Top) and Choi state (Bottom)

Using equation 3.13, we construct the dynamical map Λ_t. Note, the order of the tensor indices in figure A.5. For the dynamical map, we call the right legs first then the left legs in legList. The dynamical map Λ_t is a 64×64 matrix. We reshape this matrix into a $(2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2)$ tensor, so each leg in figure A.5 has its own dimension. This allows for easy reordering of legs.

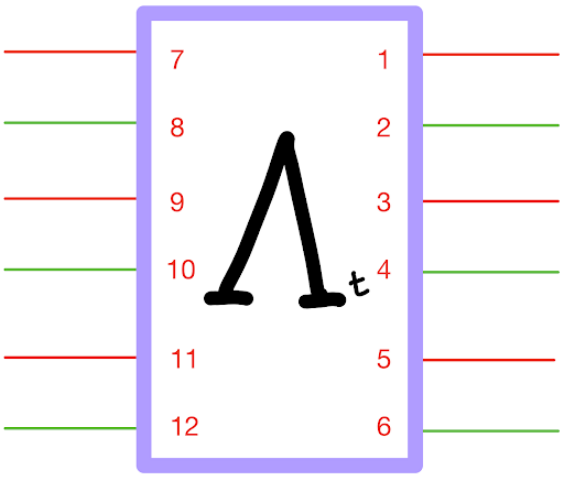


Figure A.5: Order of Λ_t indexes

We construct the two point time correlation object in figure 5.1 using equation A.1. Note, each black leg in figure 5.1

represents a red and green pair of legs in figure A.6. This call translates to figure A.6.

$$\Upsilon_{t_1,t_2} = ncon([\Psi_E, \Lambda_{t1}, \Lambda_{t2}], [[1,2,3,4], [-2,-6,5,6,7,8], [-1,-5,1,2,3,4], [-4,-8,9,9,10,10,-3,-7,5,6,7,8]]) \quad \text{(A.1)}$$

To get the separate objects Υ_{t_1} and Υ_{t_2}, we trace out the unused legs (t_1 or t_2) in $\Upsilon_{t_1\,t_2}$. For this partial trace, we connect red and green legs together, e.g. in equation A.2, the pairs of legs -3 and -7, and -4 and -8 in figure A.6 connect to trace out t_2.

$$\Upsilon_{t_1} = ncon([\Upsilon_{t_1,t_2}], [[-1,-2,1,2,-3,-4,1,2]]) \quad \text{(A.2)}$$

$$\Upsilon_{t_2} = ncon([\Upsilon_{t_1,t_2}], [[1,2,-1,-2,1,2,-3,-4]]) \quad \text{(A.3)}$$

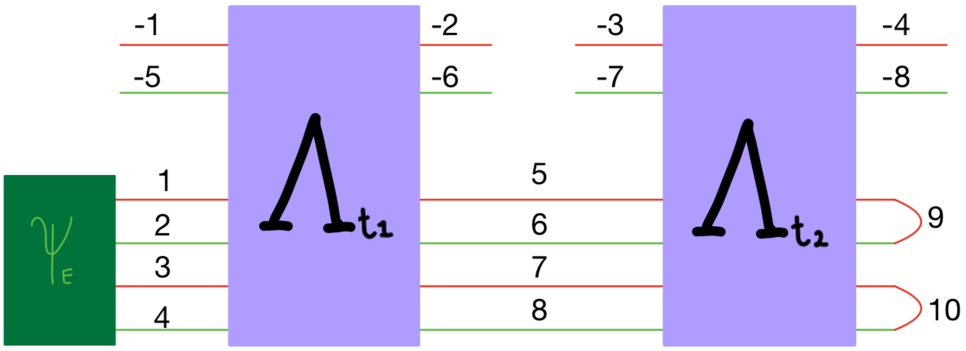


Figure A.6: Time correlation NCON call diagram